VLSI Design Internship - Task 4

RTL Design of Finite State Machines (FSM) and Control Units

Objective

Understand FSMs, Moore and Mealy machines, Traffic Light Controllers, Sequence Detectors, Verilog HDL implementation, and simulation.

Introduction to FSM

A Finite State Machine (FSM) is a sequential circuit that changes states based on clock signals and input conditions.

Moore FSM

- Problem Statement: Design a Moore FSM that cycles through three states. (S0,S1,S2)
- State Diagram: S0 → S1 → S2 → S0
- Working: The FSM transitions to the next state on every positive clock edge. The output depends only on the current state.

Mealy FSM

- Problem Statement: Design a Mealy FSM where the output becomes HIGH when the input is 1 and the current state is S1.
- Working: The FSM changes state according to the input signal. The output depends on both the state and input.

Traffic Light Controller

- Problem Statement: Design a traffic light controller using FSM concepts.
- Working: The controller cycles continuously through Red, Yellow, and Green states according to the clock signal.

Sequence Detector (1011)

- Problem Statement: Design a sequence detector to detect the binary pattern: 1011
- Working: The FSM continuously monitors the serial input stream. When the sequence 1011 is detected, the output signal becomes HIGH.

Simulation Results

The designed FSM modules were simulated using Verilog HDL and verified using waveform analysis.
Implemented modules:

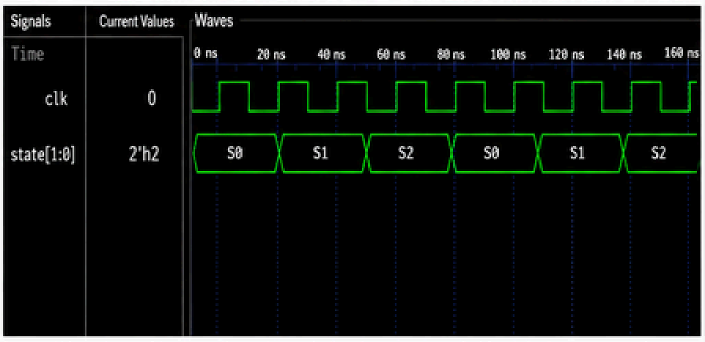
- Moore FSM
- Mealy FSM
- Traffic Light Controller
- Sequence Detector

The waveforms confirmed proper state transitions and expected outputs.

Simulation Results

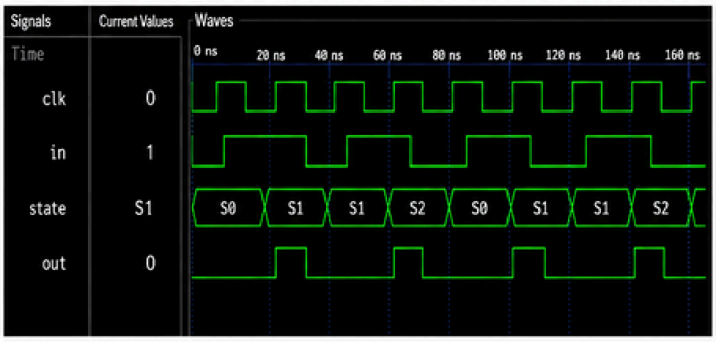
Waveforms verified correct state transitions and outputs.

1. Moore FSM (moore_fsm_waveform.png)



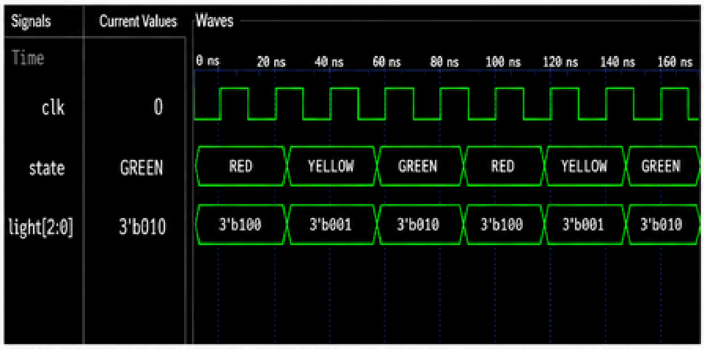
FSM cycles through three states on every clock edge.

2. Mealy FSM (mealy_fsm_waveform.png)



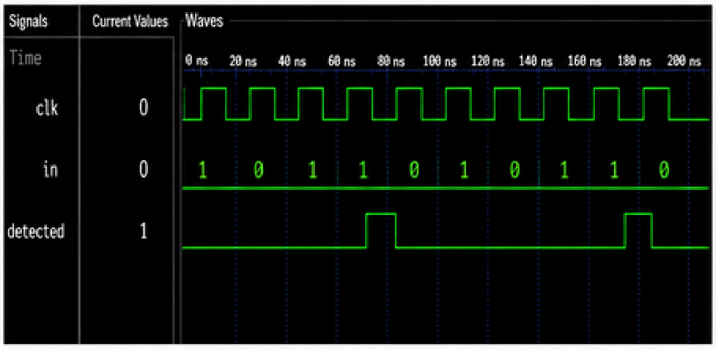
Output depends on both current state and input.

3. Traffic Light Controller (traffic_light_waveform.png)



Traffic lights cycle through RED, YELLOW, and GREEN states.

4. Sequence Detector (sequence_detector_waveform.png)



Sequence detector successfully detects pattern 1011.

Observations

- FSM transitions occurred correctly on clock edges.
- Moore FSM output depended only on the current state.
- Mealy FSM output depended on both state and input.
- Traffic Light Controller cycled through all states correctly.
- Sequence Detector successfully detected the 1011 pattern.
- Simulation results matched the expected behavior.

Conclusion

Task 4 provided practical experience in designing and verifying Finite State Machines using Verilog HDL. Moore FSM, Mealy FSM, Traffic Light Controller, and Sequence Detector circuits were successfully implemented and simulated. The task improved understanding of sequential digital design, state transitions, and control logic implementation.